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S P E C T R A V I D E O U S E R S G R O U P -- Wgtn

Newsletter Number 15 -- October 1985

The next meeting is on Monday ,October 21st at 7.30 in the Staff Common Room , Wellington Clinical School , Mein Street Newtown.

There will be a tutorial at 7.00 covering the use of OPEN & CLOSE

Following meeting is on Monday, November 18th.

Wellington SV Users Group

Secretary Darryl Alison Ph 85510 (Paraparaumu)

Treasurer Colin Chin Ph 325656 (hm)

Committee members

Garry Clark 367872

Ross Fletcher

Dennis Timmons 837200

Don Stanley 896379

Specific areas of responsibility for committee members are ...
Don -- newsletter editor/national committee ; Colin -- Treasurer
& MSX developments ; Garry -- Radio Access & public domain tape
software ; Dennis -- library ; Darryl -- program evaluation com-
mittee chairman & Kapiti users rep

Club Subscriptions

The club subscription is now \$15.00/year.

New members this month include F M Sutherland, Paul Ridgeway,
Joanne Oliver,

Note that expiry date for subscriptions is printed on the address
label of newsletter. Subscriptions not renewed a month after this
date will not receive newsletters from then.

Newsletter Contributions

If you are submitting programs to go in the newsletter they MUST satisfy the following conditions or they will NOT be printed.

1 ... be typed/printed on A4 paper with about 5 lines left blank at top and bottom of page and an inch at least around the sides

2 ... typed with a dark ribbon OR printed using emphasized print mode

3 ... include a description of what the program does , typed on A4 paper with example input

4 ... In the case of programs above 20 lines in length they should be sent on cassette , in this case there is no need for the printout as I can merge them into the newsletter. Enclose a stamped addressed envelope for return of cassette

5 ... line numbers should be continuous -- do a RENUM before printing the programs

6 ... PLEASE MAKE SURE THEY WORK . and would readers please note that I am NOT responsible for programs sent by people which do not work .

7 ... In the case of screen dumps from 3d plotters please indicate the function used and try to get the print as dark as possible -- not always easy i know .

All other newsletter contributions should be sent to P.O Box 7057 , Wellington South. They may miss being printed for a month if sent to any other box or given to anyone else. Articles should follow 1 and 2 above.

Deadline for Newsletter articles is the 30/31st of each month.

Please follow these guidelines, articles with a 4mm edge all around can not be printed , and neither can articles produced on a dot matrix printer with low resolution . They are simply unreadable after photocopying.

Correction

A couple of newsletters ago Garth had a note about storing machine code in strings and defining the routine by DEF USR=VARPTR(A\$). This should have read

```
Def Usr=256*Peek (Varptr (a$)+2) +Peek (Varptr (a$)+1)
```

Photocopying from Library

Anybody requiring articles from the library to be photocopied please send request to the LIBRARIAN, not any other committee member. We have had problems with articles not being copied when requested, please ensure that requests go to the librarian.

Also membership renewals should be sent to Box 7057, not to any other box, otherwise it is possible that the mailing list will not get updated.

Discounts for members .. All discounts offered by Einstein Scientific and Microstyle Computers do NOT include hardware. Discounts may be offered on such peripherals as floppy disks, cassettes, books, furniture, printers, monitors but NOT on SpectraVideo hardware. Individual dealers may not offer all the goods above at discounted rates, inquire from the shops just what discounts are available to club members. Note that to obtain a discount on any goods you must produce your current membership card.

GAMES SCORES

Spectron	12910	Robert Campin
Sector Alpha	1349800	Garry Clark
Kung Fu	71405	Garry Clark
	33575	Garry Clark (*)
S C Force	748600	Deirdre Chin
Telebunny	87810	Colin Chin
	26470	Don Stanley (+)
	31670	Kalima Collinson (*)
Omac	35360	Richard Bremford
	25700	Don Stanley (+)
Ninja	251350	Nicola Tuinman
Tetra	284450	Julian Stone
Turboat	68390	Garth Thornton (*)
Turboat	104610	Taane Clark
Sasa	118240	John Tuinman
Frantic Fred	21720	Rachel Hughes (level4)
Flipper Slip	31140	Kalima Collinson (level1)
Bats Alive	3540	Colin Chin (*)
Galaxy	1447	"
Jumping Jack	2600	"
TV Nightmare	7040	"
	3640	" (*)
	4980	Deirdre Chin (+)

(+) = started in normal level

(*) - started in arcade level

otherwise practice level or unknown

(Garry reports that nothing unusual happens at a score of 1

million for Sector Alpha, but by the final score he was in screen 11 and couldn't aim at anything, only move out of the way and fire aimlessly)

Palmerston North Subgroup

The Palmerston North subgroup are holding meetings on the 1st Monday of each month at COMPUTER EXPERIENCE in Palmerston North.

Club Sponsors

~~Einstein Scientific~~ -- 177 Willis Street, Wellington

MicroStyle Computers -- 29 High Street, Lower Hutt

Public Domain Software

1) Do NOT send us tapes. We will supply a standard tape which will get round the problem of people sending tapes which are too short.

2) The cost of tapes will be \$3.00 each and we will also have a charge of \$2.50 to cover handling and return postage. This is charged for each separate lot of copying done, not per tape. At present we have 4 complete tapes.

3) Any of the PD Software may also be broadcast on Radio Access in our 5 minute slot on the first Saturday of each month.

4) The tapes will contain combinations of cassette & disk programs, some may be set up for 80 columns. Adapt them as you wish to suit yourself.

5) Send orders with payment to PDS, P.O Box 7057, Wgtn South, Wellington. Make cheques payable to Garry Clark.

6) Send us your public domain software for inclusion in the software library.

CP/M public domain software has arrived from the United States Osborne group.

People wanting CPM public domain software please write to Deane Whitmore at the Taranaki Users Group who have a large range of CPM pds software. Send formatted CPM disks and a stamped addressed envelope and please give them a couple of weeks before you start moaning about not having received them. Address is 3 Lombardy Place, New Plymouth.

DOW

f people who work behind the scenes to help get the y each month. Heres some acknowledgement for the broadcasting who are so prompt with printing when newsletter , to Karen who helps me label them and to Dennis who posts them out from the hospital and s around town collecting them and returning them to ks .

K assistance line ...

hings at the rainbow ... the right things of course

s Developments

ropped in price in Australia to \$395 for the 695 for the disk drive, which makes the drive about here, but the actual computer is cheaper. According recent Tasmanian newsletter the XPress will sell for lia.

Code Corner

t up the VDP for screen 1 without using the ROM llows ::

er zero

er one

er two

```

;
; Register three
;
LD A,FF
CALL VDP1
LD A,3
CALL VDP
;
; Register four
;
LD A,3
CALL VDP1
INC A
CALL VDP
;
; Register five
;
LD A,36
CALL VDP1
LD A,5
CALL VDP
;
; Register Six
;
LD A,7
CALL VDP1
DEC A
CALL VDP
;
; Set up Name Table
;
LD B,3
LD HL,1B00 ; Start of name table from rom point of view
LD C,B1
OUT (C),L
OUT (C),H
MORE: XOR A
NEXT: OUT (B0),A
      INC A
      JR NZ,NEXT
      DJNZ MORE
;
; Now in Screen 1
;
LD A,1
LD (FE3A),A
EI
RET
VDP: OR B0
VDP1: OUT (B1),A
      RET

```

This will put the VDP into SCREEN1, rom routines such as LINE may now be used. For anyone wanting to explore graphics from CPM or without using the ROMS the above is a start point.

Don

Notes From Last Meeting

Concern was expressed by members at the lack of the treasurers report. Unfortunately the treasurer did not bother to attend either this meeting or the previous one. Once again, the treasurer has assured me that this report will be ready for insertion in this newsletter.

Treasurers Report

The treasure told me his report would be presented at the last meeting but he did not turn up. The report has not been forthcoming for this newsletter either. It is evident that the treasurer either can not or will not produce a report, and with the amount of money that the club has had go through it is unfair on all members for this to continue. This report was first requested in MARCH, perhaps it is time that the treasurer made way for someone to take over that position who has a little more time available.

Question & Answer

Does anyone know how to switch on the function key display from CPM ?

Total Computer Solutions

SVI / MSX software specialists ... P.O Box 7442, Wellington

TCS are also specialising in CONTRACT PROGRAMMING -- with over 5 years experience in SAS as well as DBASEII, PASCAL, BASIC and Z80 assembler and IBM mainframe systems (CMS/MVS/TSO) .

Watch your user group newsletter for details of titles to be released shortly for SVI/MSX systems. And send us your own titles for evaluation and possible publication .

Write to TCS, Box 7442, Wellington for further details

<<<----- Programmers / Consultants ----->>>

TCS --- Don Stanley / Garth Thornton --- TCS

Note from the EDITOR

I stated in the last newsletter that I do not any longer have the time to produce the newsletter due to work and other commitments. As no-one has expressed any interest in taking this over I can only assume that members do not consider a newsletter necessary. The number of contributions from other than myself is often small (but lately more), which I suspect has made members wary of taking on this job. Unless someone is prepared to take on the position of editor, which includes posting and labeling newsletters, this will be the last newsletter in the current format. Sorry folks, but I have been doing this virtually single-handed since June 1984 and it's time someone else did some work here.

I want to give a big thanks to those members who have supported the newsletter with articles, particularly Ross Wakelin, Elliott Young, Barry Clark, Graeme Dick, Donald Rogers and Ross Fletcher. Several others such as Deane Whitmore, Harry Paterson, John Matthews and so on have played a role with their snippets of information and programs which they have sent in. Thanks.

Once again, if anybody wants to take over the newsletter production please contact me ASAP.

TURBO PASCAL

Elsewhere in this newsletter is an advert from South Auckland Computers about Turbo Pascal. Tom Johnson has since informed me that the price of Turbo has dropped due to the high exchange rate. I don't have complete details of the new prices yet, write to Tom for details.

I have got a copy of the graphics utilities disk from Tom. Practically everything you could ever need in graphics routines is on there. However the source code for CHASE and SPACE were not on the disk.

Rom Routines

- * The routine at &H170C converts the character in the A register into uppercase. Just CALL 170C to do this.
- * The routine at &H357A is equivalent to LOCATE,,1; ie it switches on the cursor.
- * The routine at &H393E locates the cursor at the position pointed to by HL. L contains the column and H the row.
- * A completely undocumented and thoroughly useless piece of trivia is that graphics commands can have an @ character in front

DAVE

of some of the graphics positions. EG

LINE @(10,10)-(100,100),1,BF

* The following areas in ROM are not code, but rather are constants used by BASIC. This is unlikely to be all of them, but if anyones been trying to decode these areas you might as well stop.

02C9 -- 057E	Basic List of Commands
059D -- 05A7	Operator Precedence Table
05AB -- 05D6	Force Tables (used by Basic when forcing Double precision to integer and so on
05D7 -- 084B	Error messages
084C -- 0899	Constants loaded into High Memory at boot
089A -- 08A9	In / Ok / Break messages
0C27 -- 0C35	Some Tokens
13A0 -- 13B2	Redo From Start Message
1495 -- 14A5	Extra ignored Message
1DD8 -- 1DEA	Undefined Line Message
1FB1 -- 1F8E	Cassette Messages
209A -- 20CE	Cassette Prompt Messages
29E7 -- 2A14	Macro Language Command Table
2D87 -- 2DD0	Table of Notes for PLAY / Notes for Tune Register
2FAD -- 2FB0	Another Table for Play
3253 -- 3262	Table for Stick Command
33EB -- 341F	Variables sent to new bank on Switch command
3502 -- 3511	CRT Data
39B5 -- 3A14	Table of Edit Functions + Escape Codes
3E05 -- 3EAD	Keyboard Table (Used When Scanning Keyboard Every 50th Second)
4198 -- 4551	Character Set Definition Table
4CEF -- 4CF7	Used By Graphics PUT Attribute
541C -- 559B	Constants for RND/LOG/TRIG/SQRT/etc
7788 -- 779D	Device Table
7A66 -- 7B25	Initial Values for Ram (Inc Fkey String)
7D47 -- 7D8D	SV Extended Basic Message
7D8E -- 7D99	Bytes Free Message
7D9A -- 7DAF	Initialising 2nd Bank Message
7F97 -- 7FFF	Duplicate 7D47 -- 7DAF

These are correct for the Mk2 ROMS, there may be differences in the Mk1 ROMS.

* The routine at 79DC checks for a disk drive connected and searches for a disk. If found the first sector of the first track is read into locations C100 -- C100+255 and control passes there to finish entering the disk roms.

Ross

Crossword

This is a combination cryptic/normal crossword with clues going right and down as will as left and up ! Most answers have something to do with SVI/MSX !

1				2		3		4	
	5				6				
7	8			9	10				
11		12		13					
	14	15		16				17	
18		19		20	21				
	22		23		24	25			
26		27		28					
								29	
30				31	32		33		

Clues -- Across ---->

- 1 .. SV Commands to a swinging animal
- 3 .. this functions like a cabbage
- 8 .. the basis of many comparisons
- 9 .. how the mind goes after a bit of machine code
- 11 .. not the 318 or 328
- 13 .. one way you solder it , the other you drink it
- 14 .. useful device for a home computer
- 16 .. way lines move in insert mode
- 18 .. abbreviation for where we are
- 19 .. with out these devices the Z80 is worthless
- 20 .. this register keeps track of things being pushed
- 22 .. command which most SVI owners in (18) can't use
- 24 .. baby talk stretched to a Z80 command

Koss

- 26 . what we would like more of on our computers
- 28 . always comes before a sub
- 30 . link between farmer and computer ?
- 31 . what a disk user needs most

Clues across <----

- 16 . abbreviated maker of fine chips

Clues Down

- 1 .. whose doing this crossword ?
- 2 .. this is true at the end of file
- 4 .. work it out yourself !
- 6 .. connection between McDonalds & computers
- 7 .. used in OPEN statement
- 10 . logical operator with the end chopped off
- 11 . interface between user and computer
- 12 . IBM's answer to SpectraVideo
- 15 . attached to the end of (4) is fine, but on its own a headache for customs officers
- 21 . some excellent software in this library
- 22 . same as 27 up
- 25 . mid\$("spectravideo",7,2)
- 29 . first letters of one of the sounder chips around

Clues Up

- 5 .. one of two things to do in port
- 16 . same as 16 ---->
- 23 . abbreviated MOTOR ON/OFF
- 27 . same as 22 down
- 32 . first thing you see after switching on 318/328, but chopped down to function
- 33 . mathematical quantity used in circles

No prizes, but just a bizarre test of your SVI/Z80/Computer knowledge !

MSX and Spectravideo

Dear Editor:

Upon leafing through my old issues of *Creative Computing*, I had the good fortune to find the article on MSX graphics (Feb. 1984). It pays to keep old issues since one may find, in the future, something pertinent which was not of interest when the magazine was first received.

I recently purchased a Spectravideo 328 which was the machine used to develop your graphics routines. The SV 328 is a discontinued model, and I am searching desperately for peripherals (expander, parallel interface, disk drive), books, and related information.

Will the forthcoming MSX computers from Japan give the SV 328 a new lease on life—or at least provide a source of information?

As we note in "Industry Insider" this month, Spectravideo was purchased by its former manufacturing arm, Bondwell Industrial Co. of Hong Kong. Their address in the U.S. is 3300 Seldon Ct., Fremont, CA 94539. I'm not widely optimistic about the prospects of getting an expander box for the 328, as we have been trying to get one for ours for over a year with no success.

We had been hoping that the Japanese would make a commitment to MSX in the U.S., but all current indications are

that they are still playing a "wait and see" game. I don't expect to see MSX in the U.S. for at least a year—if at all.—DHA

The letter and reply to the left has been reprinted from the Sept 1985 issue of *Creative Computing*.

Also noted in the same magazine was the information that the only MSX computer in USA is the SV Express selling for \$595. The Express was shown at the CES show in the USA recently beside new releases from Atari and Commodore.

ROSS / 2/3 Commentary on Darryl

FORMATTED PRINTING OF A PROGRAM

by Graeme Dick

Have you ever tried to decipher a program (someone else's of course) in which there are multiple statements on the lines, and there are no spaces between most words? It's almost impossible to read at times, and very difficult to follow the logic of the program.

This program takes a program previously saved (to Disk or Cassette) in ASCII form and gives a printed listing in which each statement is printed on a separate line to improve the legibility. It also indents FOR/NEXT loops, and gives the option of Continuous or Paged print-out.

How does it work? Line 160 opens a file into which the program is read one line at a time. Lines 200-220 remove the line number from the string and set up the rest of the line (RL\$) for processing. Lines 230-250 remove leading spaces from the line (or a statement within a line) and then (260-300) the colon ending the statement is looked for. Allowance is made for a colon which might be embedded in a string. Lines 310-380 set up various flags (F1-F5,QF1) to test for FOR/NEXT loops, indent statements which do not have a preceding line number and to avoid indenting statements which include a 'FOR' which is not part of a FOR/NEXT loop (such as 'Open "1:" for'). Line 390 sets up the string for printing in the subroutine at line 460. If the statement to be printed is longer than 80 characters, this is handled in the loop in line 460.

Here's what the print-out would look like for two nested FOR/NEXT loops:

```
10 FOR J=1 TO 10:A=A+1:FOR K=1 TO 10:B=B+1:NEXT K:C=C+1:NEXT J
```

becomes

PROGRAM:

FOR/NEXT Loops

Page 1

```
10 FOR J=1 TO 10:
  A=A+1:
  FOR K=1 TO 10:
    B=B+1:
  NEXT K:
  C=C+1:
NEXT J
```

PROGRAM:

FORMAT.LST

Page 1

```

10 REM FORMAT.LST Graeme Dick
20 CLS:
   SCREEN,0:
   CLEAR 1000:
   PB=0:
   PL=66:
   IV$=CHR$(27)+"p":
   NV$=CHR$(27)+"q"
30 PRINTIV$"    FORMATTED PRINTING OF A PROGRAM    "NV$:
   LOCATE13,2:
   PRINT"by Graeme Dick":
   LOCATE14,3:
   PRINT"NEW ZEALAND":
   LOCATE14,4:
   PRINT"Sept 1985"
40 LOCATE0,6:
   PRINTIV$" NOTE: "NV$" THE PROGRAM TO BE PRINTED MUST":
   PRINT"    BE SAVED IN ASCII FORM"
50 LOCATE0,9:
   PRINT"The program takes a program saved in"
60 PRINT"ASCII form, and gives a printer"
70 PRINT"listing in which multiple statements"
80 PRINT"on lines are printed on separate lines"
90 PRINT"to improve the legibility"
100 LOCATE0,16:
   LINE INPUT"List (C)ontinuous or in (P)ages ? ";C$
110 C$=CHR$(ASC(C$)AND95):
   IF C$="P" THEN 120 ELSE IF C$="C" THEN 130 ELSE 110
120 LOCATE0,18:
   * INPUT"Page length (total no. of lines)";PL
130 LOCATE0,20:
   LINE INPUT"Program name ? ";NA$
140 LOCATE0,22:
   LINE INPUT"(D)isk or (C)assette file ?";H$
150 H$=CHR$(ASC(H$)AND95):
   IF H$="D" THEN H$="1:" ELSE IF H$="C" THEN H$="" ELSE 150
160 OPEN H$ + NA$ FOR INPUT AS #1
170 F2=0:
   F3=-2:
   F4=0
180 GOSUB 480
190 LINE INPUT#1,TL$:
   F1=1
200 NU=INSTR(1,TL$," ")
210 N$=LEFT$(TL$,NU)
220 RL$=RIGHT$(TL$,LEN(TL$)-NU)
230 QF=0:
   F5=0:
   RL=LEN(RL$):
   FOR J=1 TO RL:
       IF MID$(RL$,1,1)=" " THEN 240 ELSE NR=J:
       J=RL
240 NEXT J
250 RL$=MID$(RL$,NR):
   RL=LEN(RL$)
260 P=0

```

```

270 P=P+1
280 Q=INSTR(P,RL$,CHR$(34));
    C=INSTR(P,RL$,"");
    IF QF=0 THEN Q1=Q;
    QF=1
290 IF C=0 THEN C=RL
300 IF Q>0 AND Q<C THEN P=INSTR(Q+1,RL$,CHR$(34));
    GOTO 270
310 FO=INSTR(1,RL$,"FOR");
    OP=INSTR(1,RL$,"OPEN");
    TE=INSTR(1,RL$,"TO")
320 IF LEFT$(RL$,3)="FOR" THEN F3=F3+2;
    F2=0
330 IF LEFT$(RL$,2)="IF" AND FO>0 AND TE>FO AND OP=0 AND (Q=0 OR Q1>FO) THEN F3=
    F3+2;
    F2=0;
    F5=1
340 IF F3>=0 AND LEFT$(RL$,3)<>"FOR" THEN F2=1;
    IF F5=1 THEN F5=0;
    F2=0
350 IF LEFT$(RL$,4)="NEXT" THEN F2=0;
    F4=1
360 F=-(F2+F3)*(F3>0)-F2*(F3=0)
370 IF F1=1 AND F2=0 AND F3<=0 THEN PR$=N$;
    F1=0 ELSE IF F1=1 THEN PR$=N$+STRING$(F,32);
    F1=0 ELSE PR$=STRING$(NU+F,32)
380 IF F4=1 THEN F4=0;
    F3=F3-2
390 PR$=PR$+LEFT$(RL$,C)
400 GOSUB 460
410 RL$=RIGHT$(RL$,RL-C);
    IF RL$>"" AND LC<58 OR RL$>"" AND C$="C" THEN 230 ELSE IF RL$>"" THEN GOSUB 4
    50;
    GOSUB 480;
    GOTO 230
420 IF EOF(1) THEN 440
430 IF LC<58 OR C$="C" THEN 190 ELSE GOSUB 450;
    GOSUB 480;
    GOTO 190
440 CLOSE;
    CLS;
    END
450 FOR I=LC TO PL;
    LPRINT;
    NEXT I;
    RETURN
460 LL=LEN(PR$);
    LPRINT LEFT$(PR$,80);
    LC=LC+1;
    IF LL>80 THEN FOR I= 81 TO LL STEP 80;
        LPRINT STRING$(NU+F,32);MID$(PR$,I,80);
        LC=LC+1;
    NEXT I
470 RETURN
480 PG=PG+1;
    LPRINT"PROGRAM:";
    LPRINTTAB((80-LEN(NA$))/2)NA$;
    IF C$="P" THEN LPRINTTAB(70)"Page";PG ELSE LPRINT
490 LPRINT;
    LC=2;
    RETURN

```

LINE INPUT

Most of us have probably had occasion to use a routine such as

Press (some key) to Continue

to continue program execution. This normally does not appear to cause any problems, but if you use a key such as the SELECT key, and follow immediately with a line input statement, then strange things may happen. Enter the following brief program and you'll see what I mean:

```
10 SCREEN0,0:CLS:IV$=CHR$(27)+"p":nv$=chr$(27)+"q"
20 LOCATE5,0,1:PRINTIV$ " THE LINE INPUT FUNCTION "NV$
30 PRINT:PRINT"Please follow the instructions on the"
40 PRINT"screen."
50 GOSUB 1000
60 LOCATE0,6,1:LINE INPUT"LINE INPUT something: ";c$
70 PRINT:PRINT"Nothing unusual should have happened"
80 PRINT"if you pressed only the SELECT key"
90 PRINT"before the line input."
100 PRINT:PRINT"Now press a few other keys, before"
110 PRINT"pressing SELECT."
120 GOSUB 1000
130 LOCATE5,22:PRINTIV$ " Press ENTER to continue "NV$
140 LOCATE0,15,1:LINE INPUT"OOPS! What happened here: ?";c$
150 LOCATE0,19,0:PRINTIV$ " See the text of the article for the"
160 PRINT" problem, and a simple solution. "NV$
170 GOSUB 1020
180 GOTO 180
1000 LOCATE5,22,0:PRINTIV$ " Press SELECT to continue "NV$
1010 Z=PEEK(&hFD7D):IF Z<>239 THEN 1010 ELSE GOSUB 1020:RETURN
1020 LOCATE5,22:PRINTSTRING$(30,32):RETURN
```

If other keys are pressed before the SELECT key, and this will sometimes happen inadvertently, then, if a LINE INPUT statement follows, the contents of the keyboard buffer will be dumped to the screen (after the line input prompt if one was included) and the program will then wait for your line input. The "garbage" will become part of the line input.

There is a simple solution - other than not using the SELECT key in this way. Include the following line to empty the keyboard buffer after SELECT is pressed and before the line input is reached (see Newsletter No.11, June 1985). In the above program, this would go before the <Return> in line 1010.

```
defusr=&h351F:A=USR(0)
```

The same problem occurs with the CAPS LOCK and DEL keys. There may be others as well.

(by Graeme Dick)

Life, the Universe and CP/M

PART THE TWO.

Last months column has drawn some comment, some good...some bad. I thank those that took the time to comment, and remind the rest of you that this is your newsletter as well, and we need your comments to decide on content.

I have had one query in the last month, and it is so applicable that I have decided that it will be the subject for this whole column.

"How does CP/M know when to look on the second side of the double sided disk drives, and how does the hardware do it?"

This question has such far reaching implications that I will have to go right back to the basics for the answer.

A disk is a flexible piece of mylar, with a coating of some sort of magnetisable (like that word?) material. This coating is divided up (by the disk drive when it formats the disk) into a series of tracks, arranged like circles around the disk. On the Spectravideo there are forty of these tracks, numbered from 0 to 39. (have you ever noticed how computers seem to number everything from zero, and not one. It is a nuisance sometimes, but you soon get familiar with it). Each of these track is further separated into sectors, (this is where it gets tricky, so follow closely now) and on the Spectravideo, the number of tracks on each sector is not constant over the whole disk. The first track (number 0) has something like 20 sectors (I am not sure of that number cos I lost my notes), each of 128 bytes. The other tracks on the first side (side 0) have 17 sectors of 256 bytes each. On the 4000 units with the double sided drives, CP/M 2.23 puts a further 40 tracks of 17, 256 byte sectors on the other side (side 1). (following???)

Now buried inside the BIOS (remember that from last month) are the tables that tell the system how big the disks are (not 5 1/4 willy, but how much data they can hold). This table for the single sided disks says that there are 40 tracks, each track of 17 sectors, each sector of 256 bytes. But hold it right there! Didn't I just say that the first track had only 128 byte sectors?

Yep I did, and here's the rest of the story. Another parameter in those tables tells the system that the first three tracks (0,1 and 2) are not to be used for data storage. This is where the actual CP/M program resides on the disk. The first track has only 128 byte sectors because it is automatically loaded by the routines in the BASIC ROM on power up, and this routine expects the disk to have 128 byte sectors. (This is because of a hang over from the early days of disks where you could only get 128 bytes to a sector). For the double sided disk drives, it thinks there are 80 tracks of 17 sectors of 256 bytes, with the first 3 reserved for the system.

These tables also tell CP/M how many directory entries you can fit on the disk and assorted other mathematical things that come down to the fact that the smallest file you can have on a single sided disk is 1K long, even if it only has one byte in it. This is because of the way the directory is set up. CP/M cannot keep track of each individual byte in every sector of every track, so for the single sided drive it was decided that the smallest storage area (called a block) was going to be four sectors. From there CP/M looks at each BLOCK of four sectors on the disk, and numbers them from 00 up. I will leave it to you to determine the number of BLOCKS on the disk. When you save a file to the disk, CP/M searches for the next free block and in the directory entry just puts the block name (e.g. 3F) instead of putting track 17, sectors 3,4,5,6. This calculation is all done in the BDOS, from the figures given in the BIOS disk parameter tables. It might be worth mentioning that the BDOS works in 128 byte sectors, so the BIOS has to do what is called blocking and deblocking of the 256 byte sectors so that the BDOS only sees "logical" 128 byte sectors. (whew)

So, now we know how a file is defined within the directory (as blocks), and that these blocks are decoded into tracks and sectors by the BDOS. Eventually it gets down to the BDOS asking the BIOS for the data off track 37 sector 5 or somesuch.

When the BDOS asks CP/M 2.23 for a sector off tracks 40 to 79 (which are on the second side), this is what happens. When the BDOS specifies a track it wants, the BIOS subtracts 40 from it. If the result is negative, then the track must be on side 0, else what remains after the subtraction (e.g. $65 - 40 = 25$) is the track number off the second side. Remember, as far as the CP/M tables are concerned, the disk is one side of 80 tracks. If the side required is the second side (side 1), the BIOS sends a command to the floppy disk controller circuitry to set a latch, and this sends a voltage down the cable to the disk drive, which selects the head on the top side. (That's a point to note. Side 0, the most commonly used side, and the only side on single sided drives, is the bottom side when you hold the floppy so you can read the writing normally. So when you take the floppy out of the drive and drop it on the table, you are actually putting it data-side-down in the dirt..tsk tsk.)

Now, we have certainly covered a lot of ground here, but I think (hope) the it makes sense. If not let me know, and I will try again. Spot you next time.

◆◆◆◆ CLOAD JUNIOR ◆◆◆◆

JET FIGHTER Additions and Modifications

The Jet Fighter block building program appeared in the August Newsletter No.13! We have waited to this month to allow for members to respond with ideas. Only two came in and these both scroll the land below the fighter.

Garth Thornton's suggestion to speed up the joystick sensing routine in lines 640 to 710 by using an array to hold the joystick X and Y offsets is excellent. (see lines 290,300,310 and 2260 below)
His other alterations are;

DELETE line 650 to 710

```

20 DEFINT A-Z:SP=3:X=83:P=0:R=RND(-TIME)
25 GOSUB3300
290 FORK=1T08:FORJ=1T02:REM joystick
300 READA:Z(K,J)=A
310 NEXTJ,K
320 FORK=1T05:REM land colours
330 READA:GC(K)=A
340 NEXT
350 FORK=64T0191:REM land
360 LINE(0,K)-(255,K),GC(RND(1)*5+1),BF
370 NEXT
420 DEFUSR=&HC000
600 GOSUB3000
630 IFS=OTHENS=INT(RND(1)*8+1):SP=1ELSESP=3
640 X=X+SP*XZ(S,1):Y=Y+SP*XZ(S,2)
770 K=GC(RND(1)*5+1)+16-224*(CT>7):K=USR(K)
780 CT=(CT+1)MOD16
790 IFCO>10THEN500
1900 REM M/C SCROLL
1910 DATA 3A,25,F9,11,00,C1,12,21
1920 DATA 80,28,0E,10,06,08,CD,47
1930 DATA 37,DB,84,13,12,10,FA,24
1940 DATA 0D,20,F1,21,00,28,CD,3C
1950 DATA 37,21,00,C1,16,10,0E,20
1960 DATA 06,08,7E,D3,80,2C,10,FA
1970 DATA 0D,28,06,3E,F8,85,6F,18
1980 DATA EF,15,20,EA,C9
2250 REM joystick offsets
2260 DATA 0,1,1,1,1,0,1,-1,0,-1,-1,-1,0,-1,1
2270 REM land colours
2280 DATA 2,3,10,12,9
3000 PUT SPRITE1,(X-31,Y+53),4,1
3200 FORK=0T020STEP2:REM runway
3210 LINE(128-K,64)-(128-K*3,191),1:LINE(128+K,64)-(128+K*3,191),1
3220 NEXT
3230 RETURN
3300 REM M/C LOAD
3310 FORK=0T060
3320 READA$:POKE&HC(00+K,VAL("&H"+A$))
3330 NEXT
3340 RETURN

```

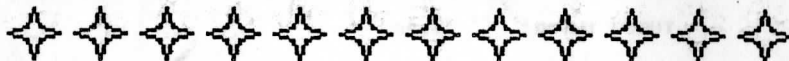
Because his Machine Code routine is not a true scroll, you cannot use any additional graphics in the lower 2/3 of the screen (y)=64) except of course for sprites. He suggests that additional sprites, either ground based guns or missile war heads, move down the screen at an angle to keep the perspective effect. Steering the Jet would obviously affect the missiles course.

REMEMBER that where lines are the same numbers as previously listed, the new ones replace the old.

FIRE BUTTON sound effect

```
762 IFM<> 0 THEN GOSUB3100
3110 SOUND7,247:SOUND8,16:SOUND11,179:SOUND12,72:SOUND13,1:RETURN
```

YOU have one more shot ! at this program. Lets hear where it can go from here, even if you don't have time to provide program lines.



Have you ever wanted to print out your own characters on your printer, but have been unable to because you don't have a downloadable printer? With the following program, simply design your character set normally using the font editor, and then run this program. It is set up to run with the modified Font Editor as in newsletter number 9.

```
10 REM SCREEN PRINTER BY ELLIOTT YOUNG
20 REM Needs to be incorporated in your own screen dump program,
30 CLEAR10000
40 CLS
50 DEFINT A-Z
60 DIM CR$(256)
70 INPUT"Filename of character set";A$:OPEN"1:"+A$ FOR INPUT AS #1
80 PRINT:PRINT"PLEASE WAIT..."
90 FOR I=1 TO 256
100 FOR X=1 TO 8:CR$(I)=CR$(I)+INPUT$(1,#1):NEXT:NEXT
110 CLOSE
120 SCREEN1,0
130 S$=STRING$(8,255):SPRITE$(0)=S$
140 PUTSPRITE0,(FF+8,DD-1):I$=INPUT$(1):CW=ASC(I$)
150 IF CW=28 THEN FF=FF+7: GOTO 140 ELSE IF CW=29 THEN FF=FF-7:GOTO 140 ELSE I
160 IF CW=31 THEN DD=DD+10:GOTO 140 ELSE IF CW=30 THEN DD=DD-10:GOTO 140
170 IF CW=8 THEN LINE(FF,DD)-(FF+8,DD+8),0,BF:FF=FF-7:GOTO 140
180 IF CW=32 THEN FF=FF+7:LINE(FF,DD)-(FF+8,DD+8),0,BF:GOTO 140 ELSE IF CW=13 T
190 DD=DD+10:FF=0:GOTO 140
180 I=CW-31
190 FF=FF+7
200 FOR X=1 TO 8:X$=BIN$(ASC(MID$(CR$(I),X,1)))
210 IF LEN(X$)<8 THEN X$="0"+X$:GOTO 210
220 FOR Q=1 TO 8:IF MID$(X$,Q,1)="1" THEN PSET(FF+Q,DD)
230 NEXT:DD=DD+1
240 NEXT:DD=DD-8:GOTO 140
```

Advt. / 1/2 Action / Action list / 3/ Action list + ? / Advt. order

TURBO PASCAL NOW AVAILABLE

Advt.....

We are pleased to announce that Borland International (USA) have just appointed us official N.Z. agents for all their products.

* This means that TURBO PASCAL (version 3.0) is now available off the shelf, direct from us - and FORMATTED for SV318/328. Price is \$240.00, but less 10% discount to all SV US members. Don't forget to quote your membership number when ordering.

* Also available are :-

TURBO TUTOR \$120

TURBO TOOLBOX \$190

* Bankcard/Visa also accepted.

* We also now have SV Graphic Utility Disc which will be supplied at only \$30.00 to all purchasers of Turbo Pascal. Anyone else interested can buy it for \$50.

* This utility enables you to find the source code for 5 demo programs - containing procedures for creative music, sprite, manipulation of sprites and collisions. High speed scrolling also possible. There is even the format of a Space Invaders Game, (enabling you to create your own).

* Explanatory notes, text files, compiler included.

* A demo disc of this utility is also available for \$6.00.

* We also stock SV 728 and the Sony Hit Bit MSX machine.

* Other services include the availability of a SV version of a Zen Assembler. Price is \$56.95. \$2.00 goes to the Auckland Club for providing the format for this utility. We supply an MSX version, but the SV version comes on the flipside.

Don't forget

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